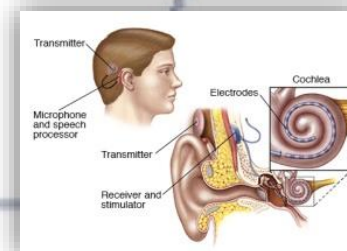
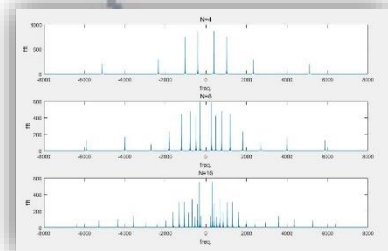
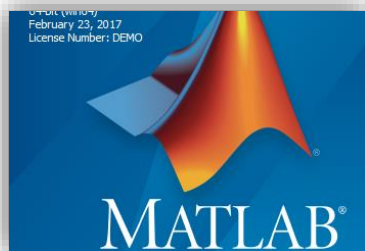


信号与系统实验

主讲人：吴光 博士

Email: wug@sustech.edu.cn



Signals and Systems (Lab)

Lab 1: MATLAB Programming

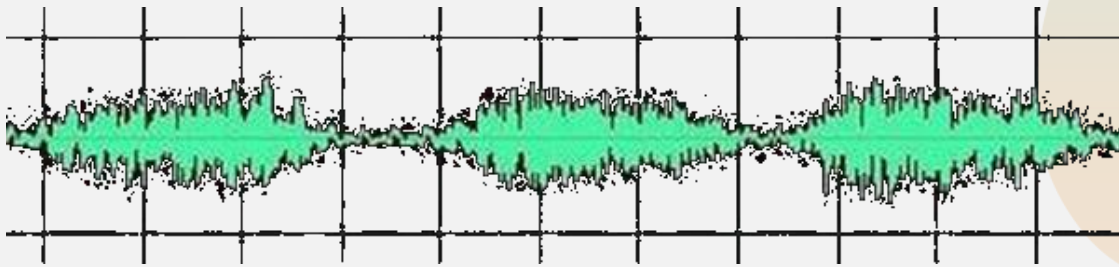
Dr. **Wu Guang**

wug@sustech.edu.cn

Electrical & Electronic Engineering
Southern University of Science and Technology

1. Objective: Analysis in frequency-domain

The music signal can be represented by two different ways.



Time-domain



Frequency-domain

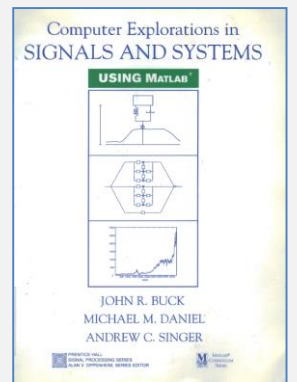


Children's Palace, Shenzhen

2.1 Labs in this course

5 Lab assignments+2 Projects

- Lab 1: MATLAB Programming (3 Weeks)
- Lab 2: Linear Time-Invariant Systems (2 Weeks)
- Lab 3: Fourier Series Representation of Periodic Signals (2 Weeks)
- Lab 4: The Continuous-Time Fourier Transform (2 Weeks)
- Lab 5: Coding Test & Low-Pass Filter (1 Week)
- Project 1: Motion detection via communication signals (3 Weeks)
- **Project 2: Modulation Classification with Deep Learning (3 Weeks)**



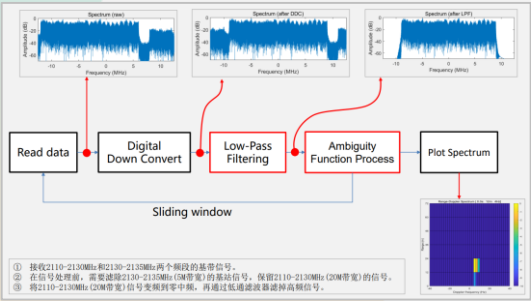
2.2 教学实施过程

- 了解实验目标
- 激发实验兴趣
- 融入思政内容

- 多径效应
- 多普勒频移
- 参考信道模型
- 监视信道模型
- 模糊函数
- 数字下变频
- 数字低通滤波器
- 距离/时间-多普勒谱



实验场景：
人体运动检测



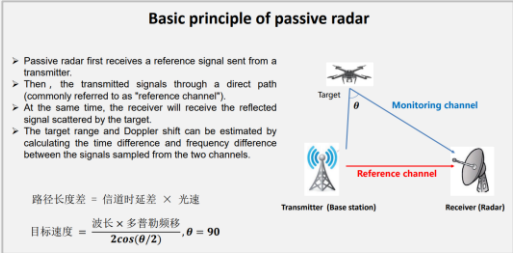
各个模块输出波形



以无人机入侵检测为例，
融入思政

原理讲解

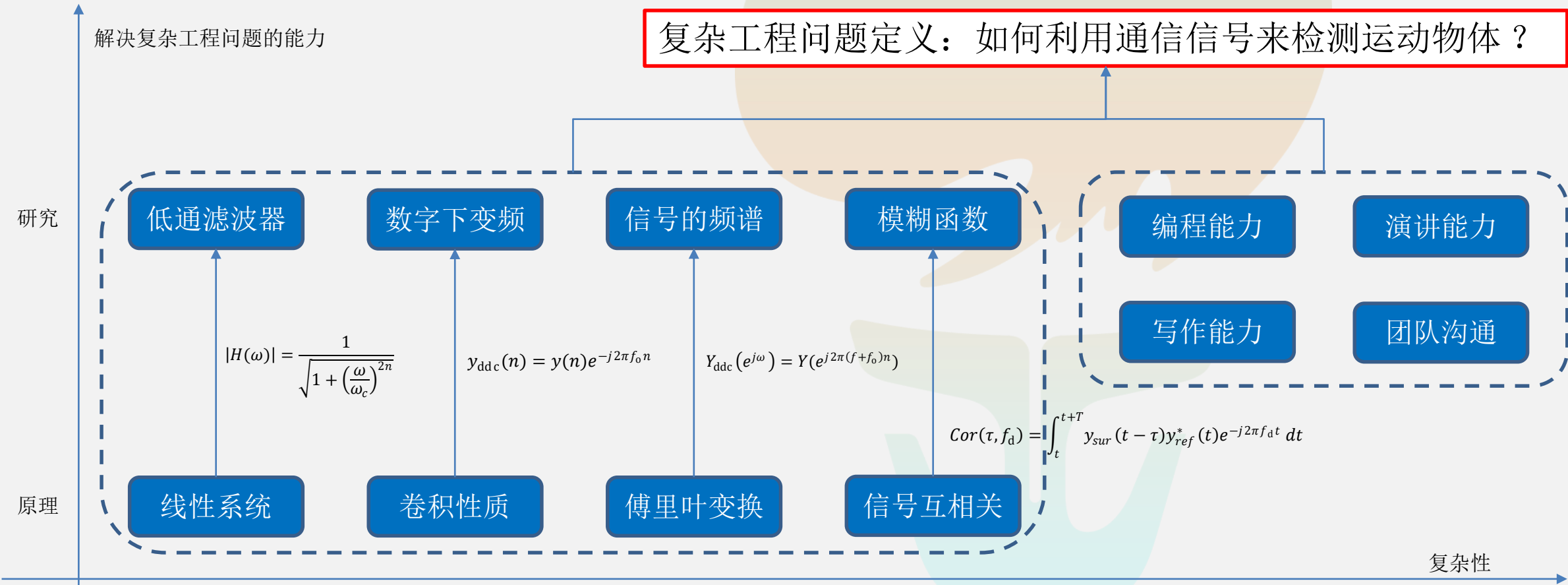
线下/线上



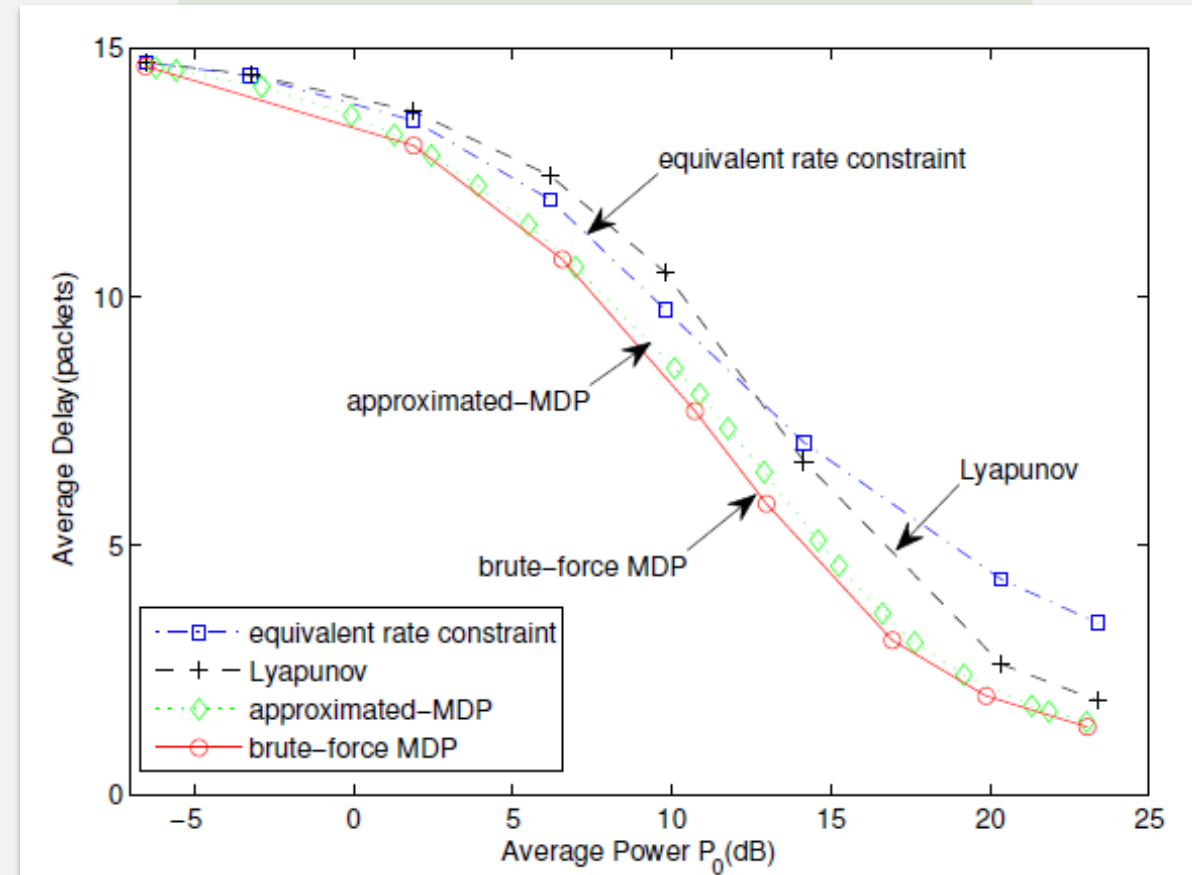
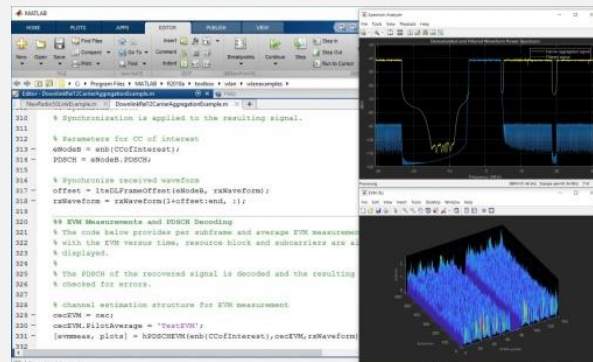
2.3 实验原理与复杂工程问题映射



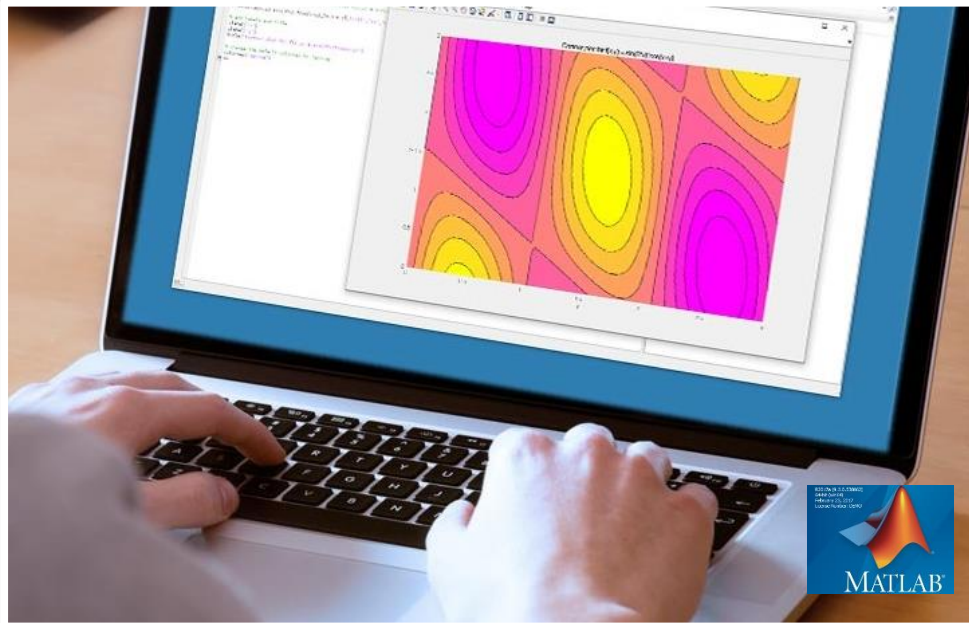
聚焦：提升学生解决复杂工程问题的能力。



What can you do with MATLAB ?



4. MATLAB (MATrix LABoratory)



- Easy-to-use, save your efforts in coding
- Everything in MATLAB is matrix
- Interpreted language
- It has its own working environment

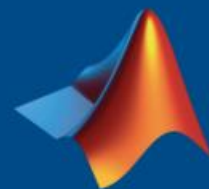
It can be used in engineering calculation, Especially for the matrix calculation.

 Computational Biology Analyze, visualize, and model biological data and systems	 FPGA, ASIC, and SoC Development Automate your workflow — from algorithm development to hardware design and verification	 Power Electronics Control Design Design and implement digital control for motors, power converters, and battery systems
 Control Systems Design, test, and implement control systems	 Image Processing and Computer Vision Acquire, process, and analyze images and video for algorithm development and system design	 Predictive Maintenance Develop and deploy condition monitoring and predictive maintenance software
 Data Science Explore data; build machine learning models; do predictive analytics	 Internet of Things Connect embedded devices to the Internet and gain insight from your data	 Robotics Convert your robotics ideas and concepts into autonomous systems that work seamlessly in real-world environments.
 Deep Learning Data preparation, design, simulation, and deployment for deep neural networks	 Machine Learning Train models, tune parameters, and deploy to production or the edge	 Signal Processing Analyze signals and time-series data. Model, design, and simulate signal processing systems.
 Embedded Systems Design, code, and verify embedded systems	 Mechatronics Design, optimize, and verify mechatronic systems	 Test and Measurement Acquire, analyze, and explore data and automate tests
 Enterprise and IT Systems Use MATLAB with your IT systems		

使用 MATLAB 和 Simulink 开展教学

全球超过 6500 所大学所采用的教学工具。

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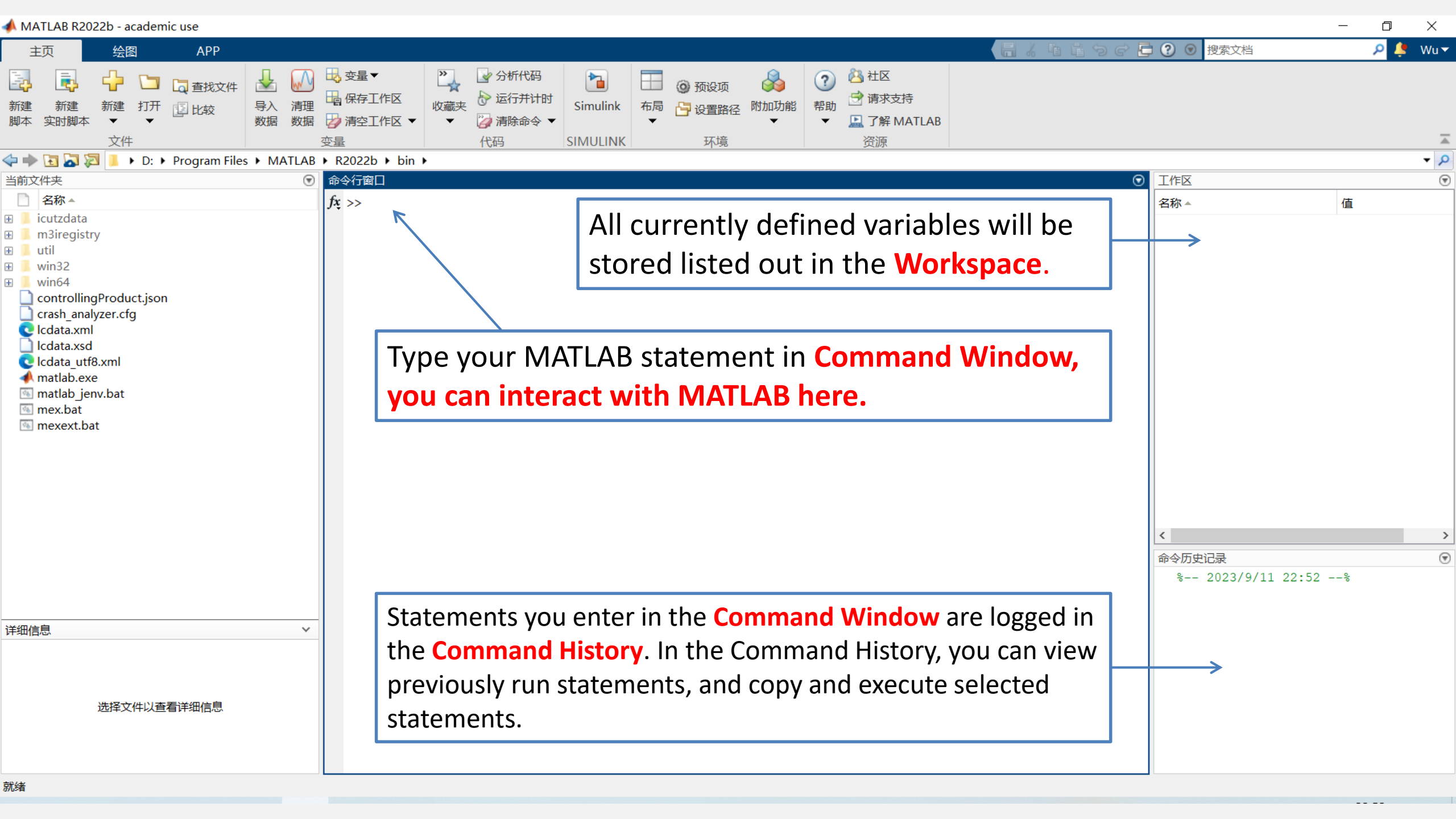
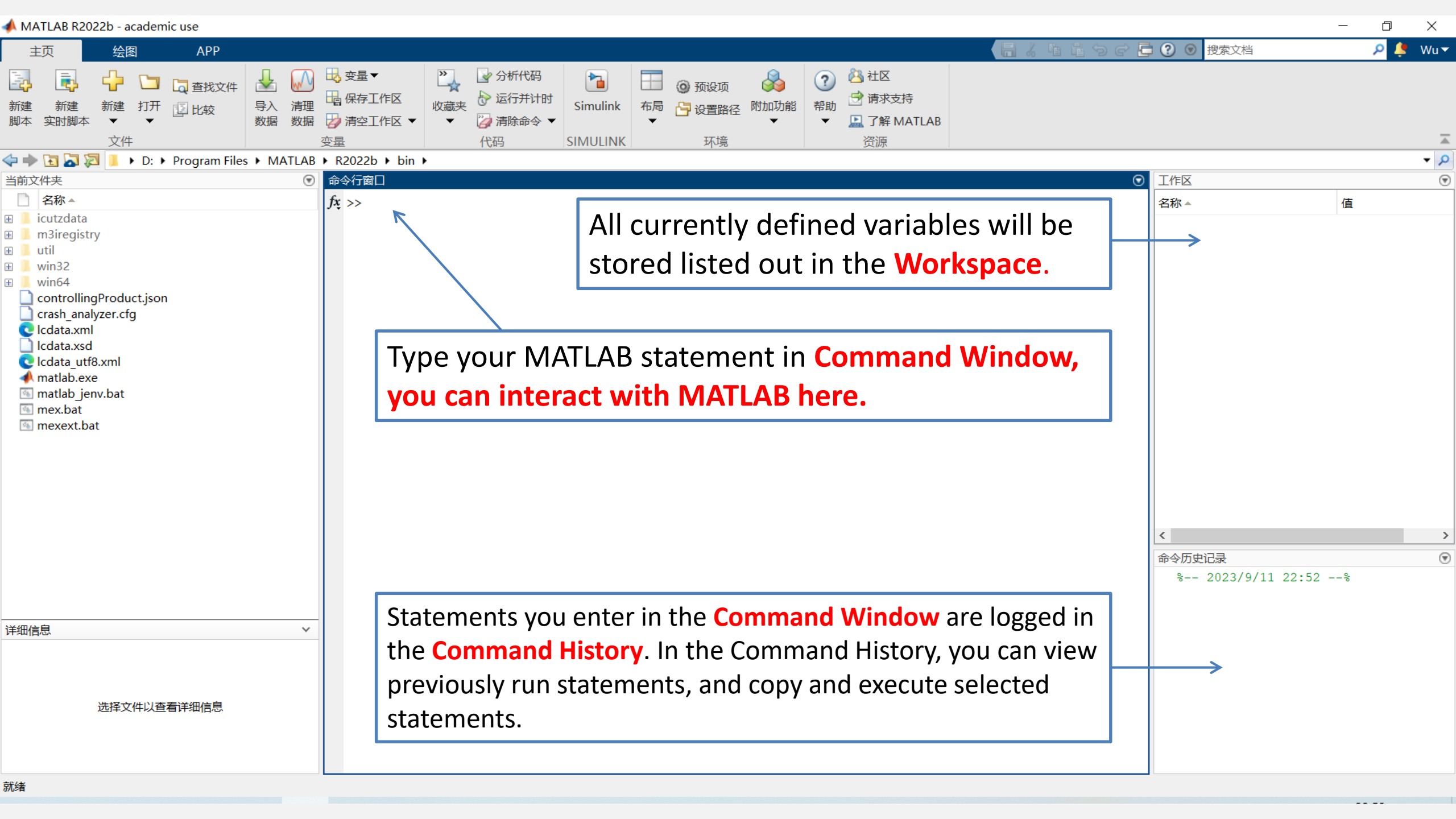
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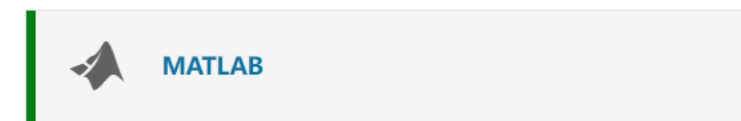
- MATLAB
- Simulink
- 5G Toolbox
- Bluetooth Toolbox
- Communications Toolbox
- Computer Vision Toolbox
- Control System Toolbox
- DSP System Toolbox
- Data Acquisition Toolbox
- Deep Learning Toolbox
- Embedded Coder
- Image Acquisition Toolbox
- Image Processing Toolbox
- Installation and Licensing
- LTE Toolbox
- MATLAB Coder
- Mapping Toolbox
- Phased Array System Toolbox
- Radar Toolbox
- Signal Processing Toolbox
- Statistics and Machine Learning Toolbox

文档 示例 函数 模块 App

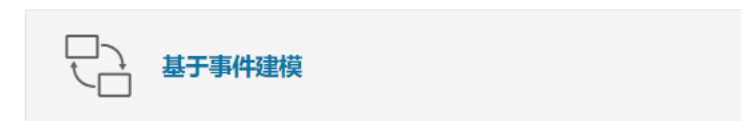
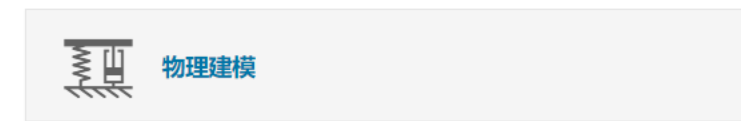
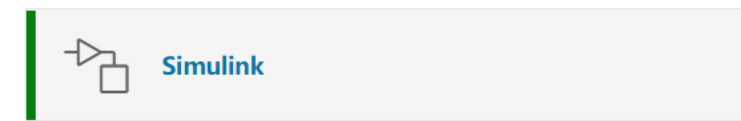
我们将持续提供翻译文档，以便您尽早看到更新的内容。
请参阅“其他版本”，获取以前版本的翻译文档。

文档

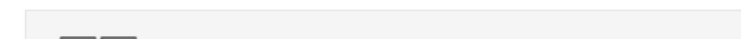
使用 MATLAB



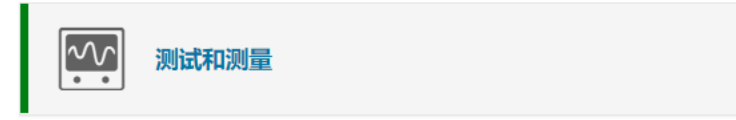
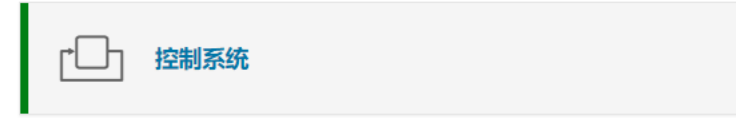
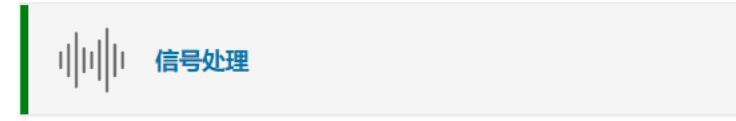
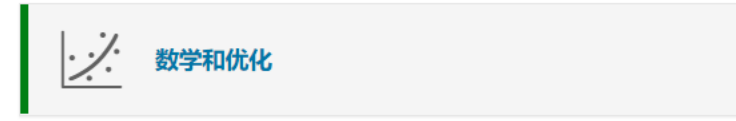
使用 Simulink



工作流



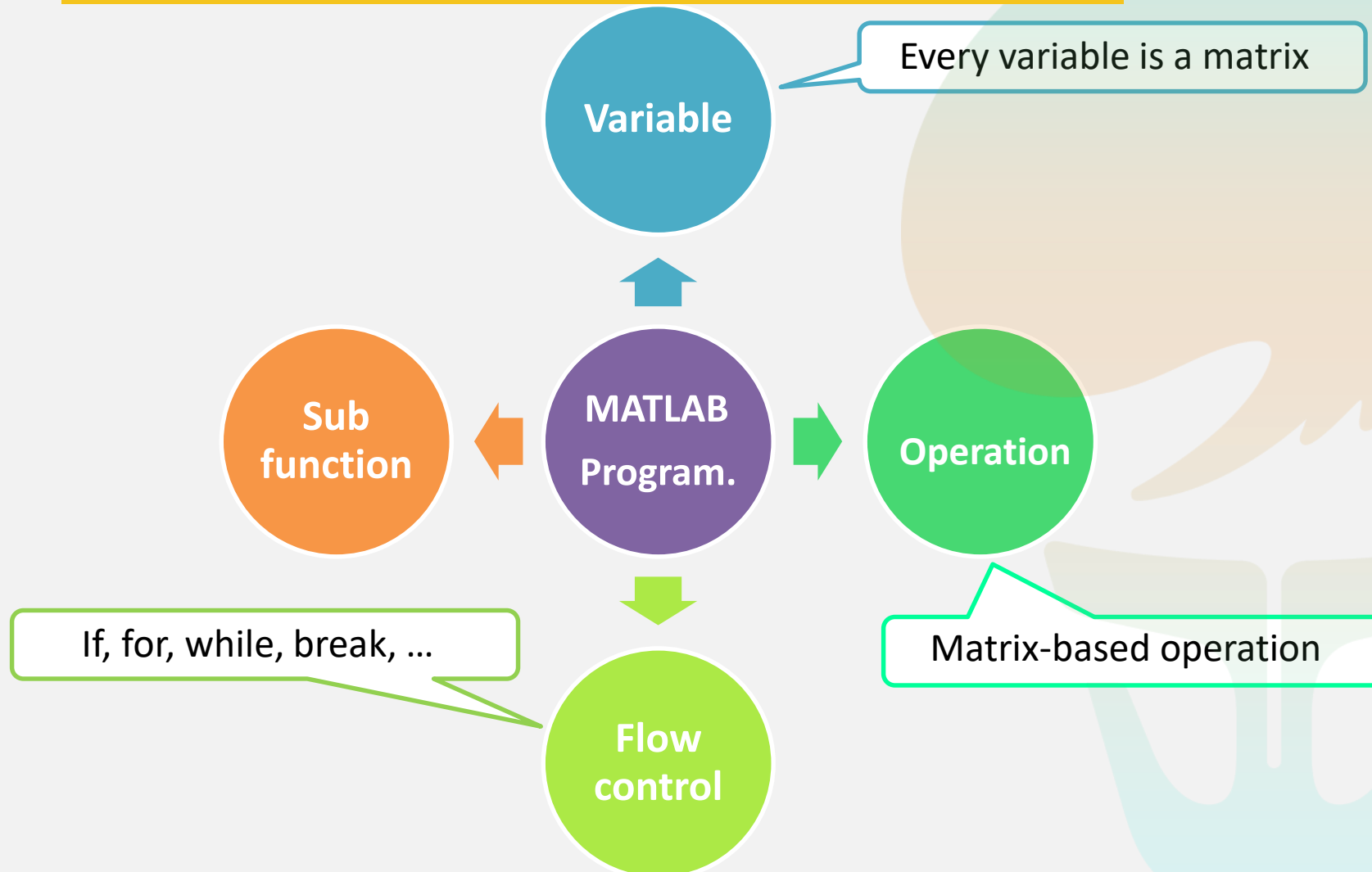
应用



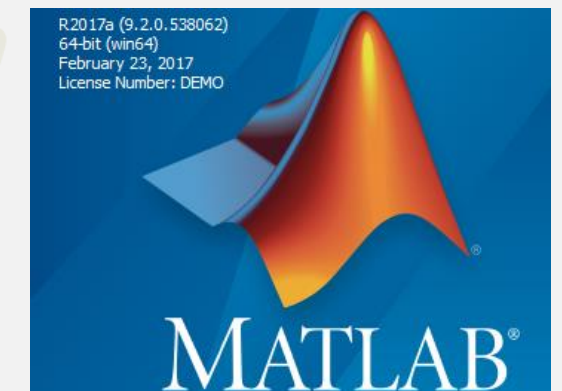
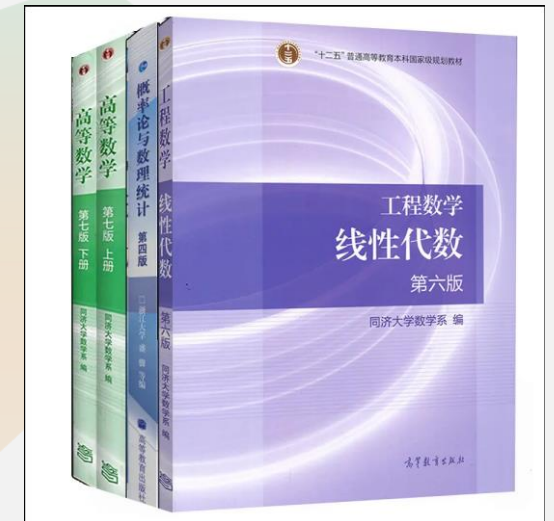
资源

- 发行说明
- 安装帮助
- 硬件支持
- Bug 报告
- 社区
- File Exchange
- 培训课程

5.1 Preliminary knowledge



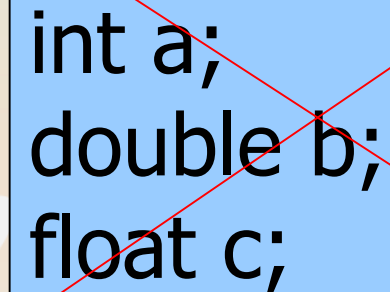
Linear algebra



5.2 How to Define Variables?

Example:

```
a=5  
a='hello world'  
a=5.5  
b=10+1i  
a=a*b  
a=a+b
```



```
int a;  
double b;  
float c;
```

No need to claim types.

- ① Latest value of variable will be stored in the workspace
- ② Semicolon (;) will stop the interaction with command window

5.3 Every Variable is a Matrix

- A vector $x = [1 \ 2 \ 5 \ 1]$

$x =$

1 2 5 1

Row Vector

- A matrix $y = [1 \ 2 \ 3; 5 \ 1 \ 4; 3 \ 2 \ -1]$

$y =$

1 2 3

5 1 4

3 2 -1

Matrix

- Transpose $z = x'$

$z =$

1
2
5
1

Column Vector

5.4 Long Vector & Matrix

t = 1:10

t =

1 2 3 4 5 6 7 8 9 10

Step=1

k = 2:-0.5:-1

k =

2 1.5 1 0.5 0 -0.5 -1

Step=-0.5

B = [1:4; 5:8]

B =

1	2	3	4
5	6	7	8

Semicolon (;) is a 'line break'

5.5 Generate vectors from functions

- zeros(M,N) MxN matrix of zeros

```
x = zeros(1,3)
```

```
x =  
    0    0    0
```

-
- ones(M,N) MxN matrix of ones

```
x = ones(1,3)
```

```
x =  
    1    1    1
```

-
- rand(M,N) MxN matrix of uniformly
distributed random
numbers on (0,1)

```
x = rand(1,3)
```

```
x =  
0.9501    0.2311    0.6068
```

6.1 Matrix Index

- The matrix indices begin from 1 (not 0 (as in C))
- The matrix indices must be positive integer

A =

3	5	3
6	8	2
2	7	3

Command ?

A =

3	5	3
6	8	2
2	7	3

```
>> A(6)
```

ans =

7

```
>> A(3,2)
```

ans =

7

```
>> A(2,:) 
```

ans =

6	8	2
---	---	---

```
>> A(1:2,2)
```

ans =

5
8

A(-2), A(0)

Error: ??? Subscript indices must either be real positive integers or logicals.

A(4,2)

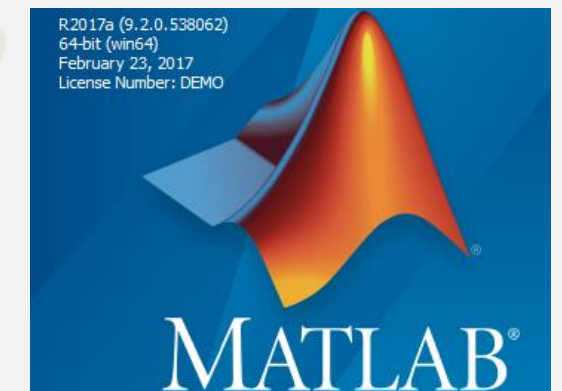
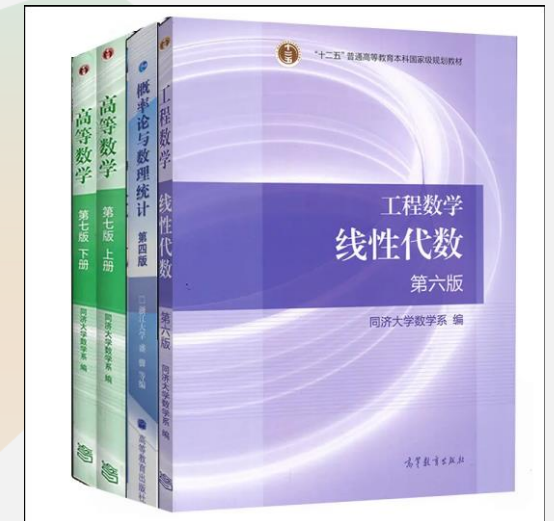
Error: ??? Index exceeds matrix dimensions.

6.2 Operators (Arithmetic)

- +** addition
- subtraction
- *** multiplication
- /** division
- ^** power
- '** complex conjugate transpose

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} * \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} & \\ & \end{bmatrix}$$

Linear algebra



6.3 Matrices Operations

Given A and B:

```
>> A = [1 2 3; 4 5 6; 7 8 9]
```

A =

1	2	3
4	5	6
7	8	9

Addition

```
>> X = A + B
```

X =

4	7	5
9	7	14
10	14	18

```
>> B = [3 5 2; 5 2 8; 3 6 9]
```

B =

3	5	2
5	2	8
3	6	9

Subtraction

```
>> Y = A - B
```

Y =

-2	-3	1
-1	3	-2
4	2	0

Product

```
>> Z = A * B
```

Z =

22	27	45
55	66	102
88	105	159

Transpose

```
>> T = A'
```

T =

1	4	7
2	5	8
3	6	9

How about

A^2

A/B

$2*A$

$A+1j*B$

$(A+1j*B)'$

6.4 Please compare

A^2 v.s. $A.^2$

$A*B$ v.s. $A.*B$

$(A+1j*B)'$ v.s. $(A+1j*B).'$

$.*$ element-by-element multiplication

$./$ element-by-element division

$.^{\wedge}$ element-by-element power

$'$ transpose

6.5 Example

```
A = [1 2 3; 5 1 4; 3 4 -1]
```

A =

1	2	3
5	1	4
3	4	-1



```
x = A(1,:)
```

x=

1	2	3
---	---	---

```
y = A(3 ,:)
```

y=

3	4	-1
---	---	----



```
b = x .* y
```

b=

3	8	-3
---	---	----

```
c = x ./ y
```

c=

0.33	0.5	-3
------	-----	----

```
d = x .^2
```

d=

1	4	9
---	---	---

```
K= x^2
```

Error: ??? Error using ==> mpower Matrix must be square.

```
B=x*y
```

Error: ??? Error using ==> mtimes Inner matrix dimensions must agree.

6.6 Useful Command--help

命令行窗口

```
>> help
```

```
--- clc 的帮助 ---
```

clc - 清空命令行窗口

此 **MATLAB** 函数 清除命令行窗口中的所有文本，让屏幕变得干净。运行 **clc** 后，您不能使用命令行窗口中的滚动条查看以前显示的文本。但您可以在命令行窗口中使用向上箭头键 **↑** 从命令历史记录中重新调用语句。

clc

打开示例

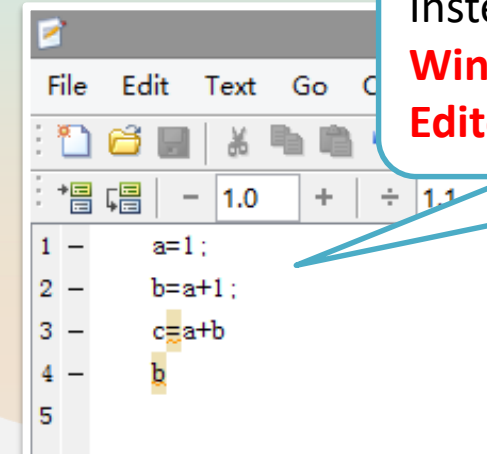
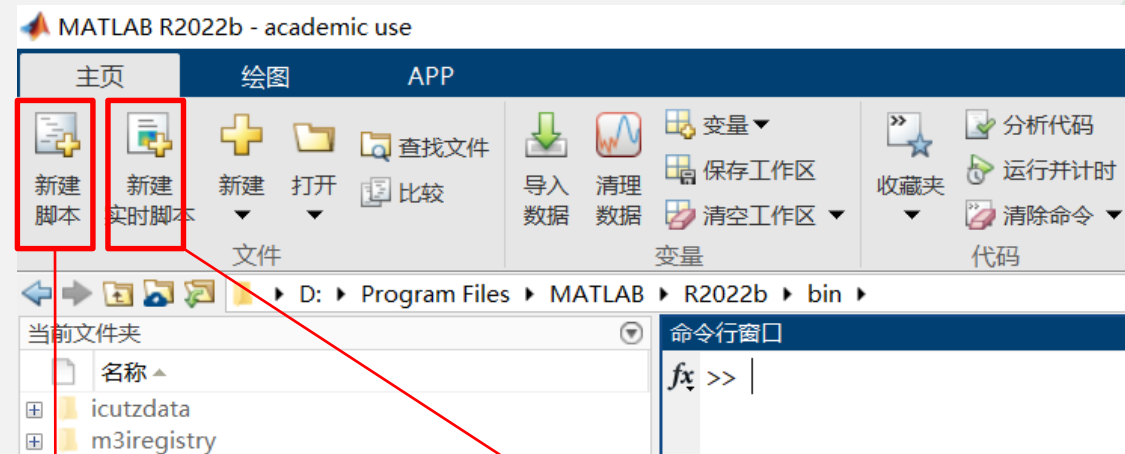
[清除所有输出](#)

另请参阅 [clear](#), [clf](#), [close](#), [home](#)

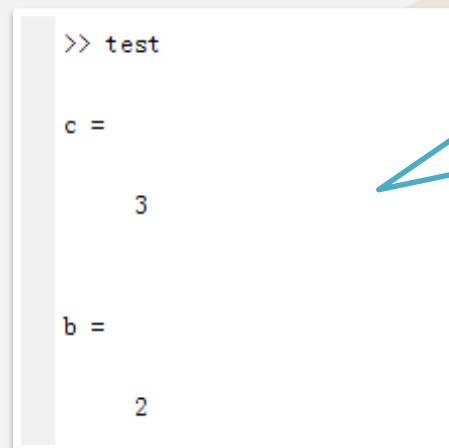
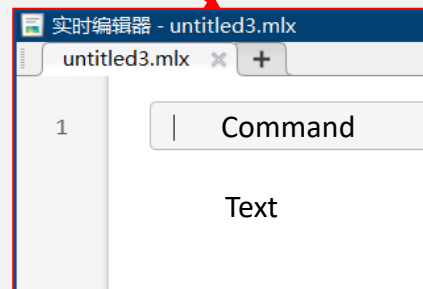
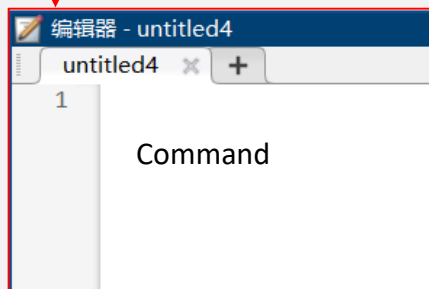
已在 R2006a 之前的 **MATLAB** 中引入
[clc](#) 的文档

fx >>

6.7 How to write a Script



Instead of typing in the **Command Window**, you can write a script in **Editor** and save in a **.m file**, like test.m.



Then, you can run the .m file in **Command Window**. The command in the .m file will be executed one by one (like a batch file).

6.8 Run and Debug



Debug line-by-line

Question ?

